SMA 2114 Revision Paper • Model Solutions

Unit: **SMA 2114** • Grounded in Prescribed Class Notes & Textbooks

QUESTION 1: Evaluate the integral of x*exp(2x) dx using integration by parts.

COMPREHENSIVE MODEL ANSWER & WORKING:

We are asked to evaluate the integral $\int x e^{2x} \, dx$ using integration by parts.

Step 1: Identify parts for integration by parts formula.
Recall the integration by parts formula:

$$
\int u \, dv = uv - \int v \, du
$$

Choose:
- $u = x$ (since derivative simplifies)
- $dv = e^{2x} dx$

Step 2: Compute du and v:
- $du = dx$
- To find v, integrate dv:

$$
v = \int e^{2x} dx = \frac{1}{2} e^{2x} + C
$$

Step 3: Apply integration by parts formula:

$$
\int x e^{2x} dx = u v - \int v du = x \cdot \frac{1}{2} e^{2x} - \int \frac{1}{2} e^{2x} dx
$$

Step 4: Simplify the remaining integral:

$$
= \frac{x}{2} e^{2x} - \frac{1}{2} \int e^{2x} dx
$$

We already know $\int e^{2x} dx = \frac{1}{2} e^{2x} + C$, so:

$$
= \frac{x}{2} e^{2x} - \frac{1}{2} \cdot \frac{1}{2} e^{2x} + C = \frac{x}{2} e^{2x} - \frac{1}{4} e^{2x} + C
$$

Step 5: Factor the expression if desired:

$$
= e^{2x} \left(\frac{x}{2} - \frac{1}{4} \right) + C
$$

Final answer:

$$
\boxed{\int x e^{2x} dx = e^{2x} \left(\frac{x}{2} - \frac{1}{4} \right) + C}
$$